

Functional Analysis Final Exam

Fall 2025

Monday December 29 9:00-12:00
Duration: 180mins

1. (15 points) Duality and Inclusions

- (1) (5 points) State the duality relations between the L_p -spaces. Which of them are reflexive?
- (2) (4 points) State the duality relations between c_0, ℓ_1 and ℓ_∞ , the space of sequences with vanishing limit, absolute summable sequences, and uniformly bounded sequences, respectively.
- (3) (6 points) State the inclusion relations between c_0, ℓ_1 and ℓ_∞ . For each inclusion, in which topology it is dense?

2. (20 points) Translation Invariant Mean

- (1) (5 points) State the Hahn-Banach norm preserving extension theorem.
- (2) (5 points) Consider the space ℓ_∞ and its vector subspace F :

$$F = \left\{ x \in \ell_\infty : \lim_{n \rightarrow \infty} m_n(x) \text{ exists} \right\}, \quad \text{where } m_n(x) = \frac{1}{n} \sum_{k=1}^n x_k.$$

Define $f : F \rightarrow \mathbb{R}$ by $f(x) = \lim_{n \rightarrow \infty} m_n(x)$. Prove that $f \in F^*$ and $\|f\|_{F^*} = 1$.

- (3) (5 points) Prove that there exists a continuous linear functional m on ℓ_∞ satisfying the properties
 - (iii) $\inf_n x_n \leq m(x) \leq \sup_n x_n$ for all $x \in \ell_\infty$,
 - (iii) $m \circ \tau = m$, where $\tau : \ell_\infty \rightarrow \ell_\infty$ is the right shift operator, i.e., $(\tau(x))_n = x_{n+1}$.
- (4) (5 points) Show that $m \notin \ell_1 = (\ell_\infty)^*$.

3. (20 points) Reflexivity

- (1) (5 points) Let E be a normed space. Define the natural embedding $E \rightarrow E^{**}$ and verify that it is an isometry.
- (2) (5 points) Let E be a reflexive space. Prove that for any $\phi \in E^*$, there exists a $x_0 \in E$ such that $|\phi(x_0)| = \|\phi\|$.

- (3) (5 points) State the Riesz Representation Theorem for continuous linear functionals.
- (4) (5 points) Show that $C([0, 1])$ is not reflexive. (Hint: Consider the linear functional $\varphi = \int_0^{1/2} - \int_{1/2}^1$.)
4. (20 points) Krylov-Bogolyubov Theorem
- (1) (5 points) State Banach-Alaoglu Theorem.
- (2) (5 points) Let $C([0, 1])$ be the space of continuous functions on $[0, 1]$. Let $T : [0, 1] \rightarrow [0, 1]$ be a continuous map from $[0, 1]$ to itself. Fix an arbitrary point $x \in [0, 1]$. For $n \geq 1$, define the empirical measure

$$\mu_n = \frac{1}{n} \sum_{k=0}^{n-1} \delta_{T^k(x)},$$

where $\delta_{T^k(x)}$ is the Dirac measure at the point $T^k(x)$ (i.e., $\delta_{T^k(x)}(\{T^k(x)\}) = 1$, $\delta_{T^k(x)}([0, 1] \setminus \{T^k(x)\}) = 0$). Prove that for any $f \in C([0, 1])$,

$$\left| \int f \circ T d\mu_n - \int f d\mu_n \right| \leq \frac{2}{n} \|f\|_\infty.$$

- (3) (5 points) Use Banach-Alaoglu Theorem to show that there exists an increasing sequence of integers $(n_j)_{j \geq 1} \subset \mathbb{N}$ and a Borel measure μ such that for any $f \in C([0, 1])$,

$$\lim_{j \rightarrow \infty} \frac{1}{n_j} \sum_{k=0}^{n_j-1} f(T^k(x)) = \int_{[0,1]} f d\mu.$$

Derive that for any $f \in C([0, 1])$

$$\int_{[0,1]} f d\mu = \int_{[0,1]} f \circ T d\mu.$$

- (4) (5 points) Deduce from (c) that μ is a T -invariant positive measure with $\mu([0, 1]) = 1$.
5. (25 points) A tale of two Hahn-Banach Theorems.
- In this problem, we prove the Hahn-Banach extension theorem using the Hahn-Banach separation theorem.
- (1) (5 points) State Hahn-Banach separation theorem (one set is open).
- (2) (5 points) Let X be a real topological vector space and $p : X \rightarrow \mathbb{R}$ a continuous sub-linear functional. Let $M \subset X$ be a linear subspace and $f : M \rightarrow \mathbb{R}$ a linear functional such that $f(m) \leq p(m)$ for all $m \in M$. Define in the product space $X \times \mathbb{R}$ the following two sets:

$$A = \{(x, t) \in X \times \mathbb{R} : t > p(x)\}, \quad B = \{(m, f(m)) : m \in M\}.$$

Show that A and B are convex, A is open in the product topology of $X \times \mathbb{R}$ (which is a topological vector space), and $A \cap B = \emptyset$.

- (3) (5 points) Apply the Hahn-Banach separation theorem to obtain a nonzero continuous linear functional $\Phi : X \times \mathbb{R} \rightarrow \mathbb{R}$ and $\alpha \in \mathbb{R}$ such that

$$\Phi(b) \leq \alpha < \Phi(a) \quad \forall a \in A, \forall b \in B.$$

Show that $\alpha = 0$.

- (4) (5 points) Show that Φ can be written as $\Phi(x, t) = \varphi(x) + \beta t$ for some continuous linear $\varphi : X \rightarrow \mathbb{R}$ and $\beta \in \mathbb{R}$. Prove that $\beta > 0$.
- (5) (5 points) Let $\tilde{f}(x) = -\frac{1}{\beta}\varphi(x)$. Prove that $\tilde{f}$ is a continuous linear functional satisfying $\tilde{f}|_M = f$, and $\tilde{f}(x) \leq p(x)$ for all $x \in X$.
6. (10 points (bonus)) (1) (5 points) Let $L_2([0, 1])$ be the L_2 -space on the interval equipped with the Lebesgue measure. For $\phi \in L_2([0, 1] \times [0, 1])$, define the operator

$$T_\phi(f)(x) = \int_0^1 \phi(x, y)f(y) dy.$$

Prove that $T_\phi : L_2([0, 1]) \rightarrow L_2([0, 1])$ is bounded. $\|T_\phi\| \leq \|\phi\|_2$.

- (2) (5 points) Prove that $T_\phi : L_2([0, 1]) \rightarrow L_2([0, 1])$ is compact. (Hint: ϕ can be approximated by a sequence of functions in the form $\phi_n(x, y) = \sum_{i=1}^{k_n} g_{n,i}(x)h_{n,i}(y)$ where for each n , the summation is finite and $g_{n,i}, h_{n,i} \in L_2([0, 1])$. What type of operator is T_{ϕ_n} ?)